Pre-Class Assignment: Chapter 3

The Value of Prediction

ECON 470 — Microeconomics of Artificial Intelligence

Spring 2026

Reading: Gans, Chapter 3 (The Value of Prediction)

Instructions: Answer each question in 1–3 sentences. These questions are designed to be straightforward if you did the reading.

You must submit this assignment in person. The assignment must be handwritten. No photocopies will be accepted.

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1. In the 2×2 model, an agent chooses an action $a_i \in \{1, 2\}$ and there are two possible states $\theta \in \{1, 2\}$. What payoff does the agent receive when their action matches the state? What payoff do they receive when it doesn't?
2. Without any signal, the agent chooses the action that maximizes expected payoff given their prior. Since $p > \frac{1}{2}$, which action do they choose and why?
3. Gans defines V as the value of perfect information. In words, what does V measure? Write the formula.
4. In Section 3.1.1, the signal s is correct with probability e and incorrect with probability $1-e$. What condition must hold for the agent to prefer following the signal over ignoring it?
5. In the 2×2 model, the value of (perfect) information is $V = (1-p)(R-r)$. What happens to V when p increases toward 1? What is the intuition?
6. What is an ROC curve? What do the two axes represent?
7. In Figure 3.1, what does the point where the indifference curve is tangent to the ROC curve represent?
8. The willingness to pay for a prediction of quality Λ is $V(1) = (1-p)(R-r)(\Lambda - 1)$. What are the two factors that determine how much an individual agent is willing to pay?
9. Gans argues that the statistical loss function used to train a machine learning model should be connected to the decision context. Why is squared error loss not always appropriate?
10. In one sentence, state the key takeaway of Chapter 3.